



Dynamics of the global trade in live reptiles: Shifting trends in production and consequences for sustainability



Janine E. Robinson ^{*}, Richard A. Griffiths, Freya A.V. St. John, David L. Roberts

Durrell Institute of Conservation and Ecology, School of Anthropology and Conservation, Marlowe Building, University of Kent, Canterbury, Kent CT2 7NR, UK

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ABSTRACT

Biodiversity-rich countries provide wildlife for the exotic pet trade, but the implications of this for conservation, sustainable use and livelihoods remain poorly understood. CITES Appendix II import data from 1996 to 2012 were used to analyse spatial and temporal trends in live reptiles, a group comprising a substantial component of the commercial wildlife trade. Between 2001 and 2012 the trade declined by a third. The decrease was greatest in wild-caught reptiles (70%), but imports in captive-bred reptiles also decreased (40%), due to reduced trade in green iguanas. Imports originating from captive sources comprised about half of the total trade over the period. In contrast, there was a nearly 50-fold increase in imports of ranched reptiles, dominated by royal pythons from sub-Saharan Africa, but including a recent upsurge of ranched turtles from South America and Asia. Additionally, the proportion of reptiles sourced from 'range countries' (where species naturally occur in the wild) declined. Numbers of reptiles captive-bred within consumer countries to supply domestic markets are difficult to obtain, but may be impacting international trade. Captive breeding may ease collection pressure on wild populations, but might also divert benefit flows, impacting local livelihoods. Ranching may benefit livelihoods and have low impacts on natural populations, but along with captive breeding, could be detrimental if loopholes allow wild animals to be exported as ranched. Given the shift from wild to ranched reptiles, more information is required on the benefits and impacts of commercial ranching operations for traded reptile species.

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1. Introduction

Overexploitation contributes significantly to the extinction risk of threatened species (Rosser and Mainka, 2002). If sustainability is an ultimate conservation goal, it is crucial to understand supply, demand and the temporal and spatial dynamics of resource use. Millions of live animals and plants, as well as their parts and derivatives, are traded each year to supply consumer demand around the world (Broad et al., 2003; Nijman, 2010; Smith et al., 2009). The legal international trade in wildlife, excluding fisheries and timber, was estimated to be worth ~\$24 billion in 2005 (Engler and Parry-Jones, 2007) but domestic and illegal trade is much more difficult to value. Wildlife trade can impact species conservation (O'Brien et al., 2003; Shepherd and Magnus, 2004; van Balen et al., 2000) but also has important social and economic implications (McNeill and Lichtenstein, 2003; Roe, 2002, 2008). The Convention on International Trade in Endangered Species (CITES) was established to help regulate international trade in wild species and ensure it does not threaten their survival. Over 35 000 species

are afforded varying levels of protection through CITES, according to their conservation status and risk from trade, which is regulated through an import–export permitting system.

In some cases, captive breeding may provide a suitable alternative to wild collection (Jepson et al., 2011). It can reduce pressure on wild populations, and captive-bred animals are subject to less stringent CITES controls than wild-sourced animals. Indeed, many commonly kept pets are bred in consumer nations, and this can contribute to research regarding biology, husbandry and disease (Marano et al., 2007). However, captive breeding can also reduce benefit flows to countries where species originate, counter to the Convention on Biological Diversity which recognizes sovereign rights of states over their natural resources and advocates “fair and equitable sharing of benefits arising out of their use” (CBD, 2014). Captive breeding may also disconnect suppliers from source habitats thus limiting opportunities for sustainable use and conservation initiatives. Additionally, captive breeding has been linked to laundering of illegally wild-caught animals (Brooks et al., 2010; Lyons and Natusch, 2011), demonstrating the complex and varied nature of the benefits and impacts of alternative production strategies for supplying the trade.

^{*} Corresponding author. Tel.: +44 (0)1227 827 056x4076.

E-mail address: jr418@kent.ac.uk (J.E. Robinson).

'Ranching', defined within CITES as "rearing in a controlled environment of specimens which have been taken as eggs or juveniles from the wild, where they would have a very low probability of surviving to adulthood" (CITES, 2014a), is another system used for producing reptiles. Ranching relies on harvesting young life stages that regularly experience high mortality in the wild, and is therefore considered a relatively benign method of exploitation (Hutton et al., 2001). In some cases, the harvesting is compensated for by the release of some offspring back into the wild. Ranching is practiced within the country where the species occurs, and if well managed, has potential to benefit both livelihoods and conservation (Gordon and Ayiamba, 2003; Moyle, 2013; Thorbjarnarson, 1999).

A substantial component of the international wildlife trade is made up of reptiles and their products, (Caldwell, 2012; Hoover, 2000; Kasterine et al., 2012). For example, excluding caviar extract, the trade in reptiles accounted for 84% of the value of CITES-listed animals and animal products imported into the EU in 2010, mostly consisting of leather products and skins (UNEP-WCMC, 2012). Additionally, commercial and recreational breeding of reptiles has increased in recent years (Auliya, 2003). The live reptile trade into the EU was valued at \$4.3 million in 2010, accounting for 22% of all live imports by value, superseded only by mammals (UNEP-WCMC, 2012).

Some reptile groups, particularly those associated with freshwater and marine habitats are facing disproportionately high extinction risks (Böhm et al., 2013). These risks are particularly severe amongst turtles, terrapins and tortoises, which in addition to suffering a range of threats (Bugoni et al., 2001; Lewison and Crowder, 2007), are traded extensively as food, curios, pets and use in traditional medicine (Gibbons et al., 2000; Nijman and Shepherd, 2007). Further, there is often limited information about the viability of wild populations (Pough, 2013). Reptiles with small ranges and narrow niche requirements are particularly vulnerable to anthropogenic threats (Böhm et al., 2013). Equally, turtles, snakes and crocodilians that have life histories with prolonged adult survival are vulnerable to commercial exploitation (Pough, 2013).

There is a lack of information regarding the number of live reptiles in trade, where they come from, and the production strategies used to supply them. Using CITES Appendix II trade data we review trends in the production strategies used to supply live reptiles for commercial international trade from 1996 to 2012. In particular, we address the following questions: (1) How has the relative importance of captive-bred, ranched, and wild-sourced animals changed over time? (2) How have changes in production strategies been reflected in global supply routes? (3) What are the consequences of the temporal and spatial dynamics for long term sustainability of the live reptile trade? We focus on the commercial trade in live reptiles, which are predominantly destined for pet trade, but in some cases supply production industries such as farming for meat and skins. Whilst the CITES Trade Database provides substantive data on trade in endangered species at a global scale, we acknowledge that it represents a subset of the entire global trade in reptiles as it does not take into account non-CITES species, illegal or unreported trade, and trade conducted within national borders.

2. Methods

The CITES Trade Database is managed by the United Nations Environment Program and World Conservation Monitoring Centre (UNEP-WCMC) on behalf of the CITES Secretariat with data collated from CITES annual reports submitted by the Parties (CITES signatories). Data on all live reptiles traded globally since 1996

were supplied by UNEP-WCMC [7 April 2014]. Data up to and including 2012 were chosen for analysis based on the completeness of the annual report submission record [dated 4 April 2014].

Ambiguous trade records, such as live specimens traded with units in 'kg' or 'shipments' (representing 0.3% of the data set), rather than as whole units, were excluded. Only data on Appendix II (representing 92.3% of all live reptile trade), commercially traded (coded purpose 'T' in the Trade Database – representing 99.2% of Appendix II reptiles) were analysed. Quantities of reptiles reported by importing countries were analysed rather than quantities reported by exporting countries because some exporting countries report the number of permits issued rather than actual numbers of reptiles exported (UNEP-WCMC, 2013). Only direct trade between exporting and importing countries was analysed, re-export data were excluded. This was because the inclusion of re-export data can lead to double counting and therefore elevated trade records. Re-export data are also unrelated to the country of origin of traded specimens. Because the 27 EU members are not required to report within-EU trade (due to the free trade agreement), rarely reported within-EU trade records were removed.

Comparisons were made between captive-bred (source 'C'), ranched ('R') and wild ('W') reptiles. Remaining sources including first generation ('F'); confiscated or seized ('I'); pre-convention ('O'); unknown ('U') and source unreported ('blank') are incorporated in the 'total trade' figures that are presented.

Genus level records were not included when reporting the number of different species imported over time. However, these records were not excluded for any other analysis. Following CITES standard nomenclature as adopted by the Conference of the Parties (CITES, 2014b), and to avoid duplication, any data reported using synonyms were combined with data reported under the accepted name. Data recorded at subspecies level were combined with species data.

Exporting countries were assigned to geographical regions following the ISO 3166 list of countries maintained by the International Organisation of Standardization and used by the IUCN Red List of Threatened Species. For the top 100 species in trade, exporting countries were categorised as 'range' and 'non-range' according to species range information on the UNEP-WCMC SPECIES+ database and the IUCN Red List. Countries where species were listed as 'introduced' were considered 'non-range'. To compare imports from range versus non-range countries, the percentage of trade coming from range countries in 1996 was compared with 2012, excluding species which were CITES listed post-1996.

Analyses were conducted using IBM SPSS Statistics 20. Pearson's correlations were used to test for relationships between the proportions of imports from different sources (captive-bred, ranched, and wild) over time. A Sign Test was used to test for differences between proportions originating from range versus non-range countries across species.

3. Results

3.1. Global reptile trade

Over 18.8 million live Appendix II reptiles were imported globally for commercial purposes between 1996 and 2012. Most (96.8%) were captive-bred, ranched or sourced from the wild, with remaining sources (D, F, O, I, U and 'blank') accounting for 3.2%.

Following peaks in 1996 and 2001, there has been an overall decline in annual Appendix II reptile imports. Imports decreased by 32.8% from 2001 to 2012 at an average rate of 3.4% per year (Fig. 1a).

Green iguana (*Iguana iguana*) consistently dominated the live reptile trade, with annual imports peaking at nearly one million

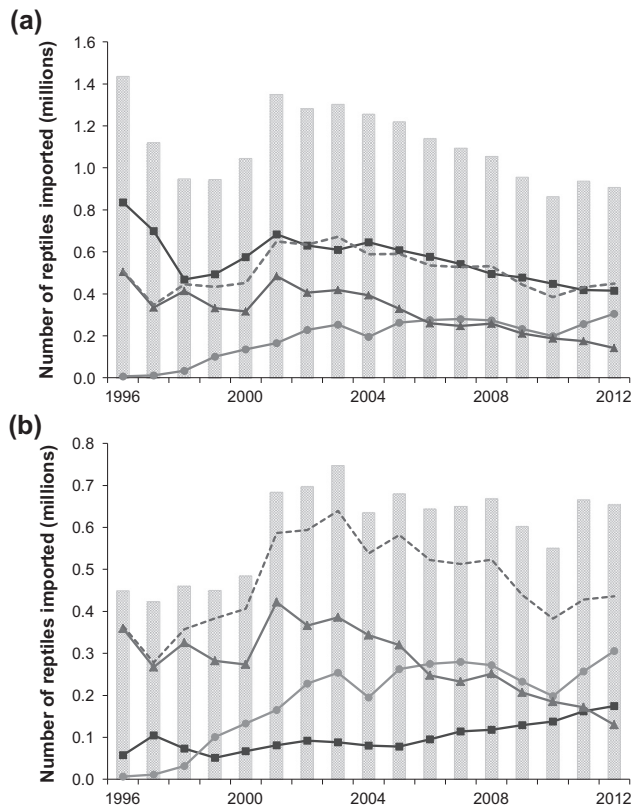


Fig. 1. (a) Global imports in CITES Appendix II live reptiles traded for commercial purposes from 1996 to 2012 displaying totals in captive-bred (solid line with square markers), ranched (solid line with round markers), wild (solid line with triangle markers) and ranched plus wild (dashed line) against total annual imports (bars). When wild and ranched imports are combined, imports follow a similar pattern to that of total trade. (b) As above, excluding the green iguana (*Iguana iguana*), which accounts for 46.2% of all trade.

in 1996 (Fig. 2). In total, 8.7 million *I. iguana* were imported, accounting for close to half (46%) of all imports between 1996 and 2012. However, *I. iguana* imports decreased by 62% from 2001 to 2012. If *I. iguana* is removed from the data, global imports increase to 2003 and then decrease by 12% from 2003 to 2012 (Fig. 1b). The royal python (*Python regius*) was the second most highly traded reptile, with about 2.7 million imported between

1996 and 2012, accounting for 14% of total trade. No other species comprised more than 4% of total trade, and the top ten most imported species accounted for 77% of all trade (Table 1).

From 1996 to 2012, the number of Appendix II species imported for commercial purposes increased from 142 to 234, totalling 388 different species over the entire time period, with 119 species common to both 1996 and 2012. This excludes 15 genus level records in 1996 and five in 2012, the decrease of which is likely to be a result of improved reporting. A total of 54 species in the dataset were listed post-1996, including 35 of the 234 species in trade in 2012.

3.2. Trends in live reptile production systems

Consistent with the decrease in total Appendix II reptile imports, there has been a decrease in the number of wild caught reptiles imported globally, decreasing by 71% from 2001 to 2012. Meanwhile, imports of ranched reptiles increased by more than 4740% from 1996 to 2012. Imports of captive-bred reptiles decreased by 39% from 2001 to 2012. However, if *I. iguana* is excluded, the trend shows a 202% increase in imports of captive-bred reptiles between 1996 and 2012 (Fig. 1b) indicating that this species is responsible for the general decrease in captive-bred reptile imports.

Captive-bred reptiles represented about half of annual trade from 1998 to 2012, after peaking around 60% in 1996 and 1997. Wild reptile imports decreased from 35% in 1996 to 16% in 2012. Meanwhile, ranched reptiles increased from 0.4% to 34% of total trade. There was a significant negative relationship between the proportion of wild imports and the proportion of ranched imports between 1996 and 2012 ($r = -0.85$, $n = 17$, $p < 0.001$) and between the proportion of captive-bred and ranched imports ($r = -0.76$, $n = 17$, $p < 0.001$), demonstrating that trade in ranched reptiles increased as trade in both wild and captive-bred reptiles decreased.

If however, *I. iguana* and *P. regius*, which dominate the trade in captive-bred and ranched imports (Table 1), are removed, an alternative pattern emerges. In this case there is a significant positive correlation between the proportion of captive-bred and ranched imports ($r = 0.86$, $n = 17$, $p < 0.001$), as both increased over the time period. Conversely there is a significant negative relationship between the proportion of wild and captive-bred imports ($r = -0.96$, $n = 17$, $p < 0.001$) and the proportion of wild and ranched imports ($r = -0.96$, $n = 17$, $p < 0.001$), indicating that captive-bred and ranched imports increased as wild imports decreased.

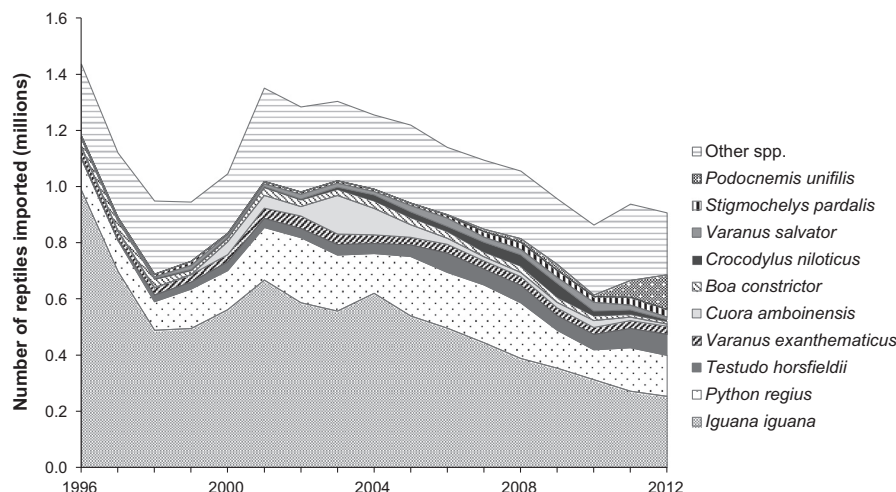


Fig. 2. Summary of trade in ten CITES Appendix II reptile species most imported for live commercial trade between 1996 and 2012, including all sources. This graph displays stacked totals and not cumulative totals; therefore the thickness of the area indicates the quantity of trade for each species.

Table 1

Summary of trade in live CITES Appendix II reptiles most imported for commercial purposes as captive-bred, ranched and wild between 1996 and 2012, including the percentage trade each species comprised for each source over the time period, and the percentage change in the number of imports of each species from 1996 to 2012. Percentage change was calculated between 1996 and 2012 except in the following cases where trade records were incomplete: *C. amboinensis* (2000–2012), *A. cartilaginea* (2005–2012), *P. mucosus* (2000–2012), *G. elegans* (1997–2012), *C. niloticus* (1997–2012), *P. unifilis* (2005–2012), *T. horsfieldii* (2001–2012), *H. grandis* (2006–2012), *H. annandalii* (2009–2010), *P. sebae* (2006–2010), *T. graeca* (2005–2012), *C. amboinensis* (2000–2012), *A. cartilaginea* (2006–2012) and *U. dispar* (1997–2012). Bold = species listed as threatened on the IUCN Red List (Endangered, Vulnerable or Critically Endangered).

Source	Total trade (mean per annum $\pm$ se)	Top 15 species in trade (% of total trade 1996–2012, % change 1996–2012)
All sources	18858195 (1109306 $\pm$ 42113)	<i>Iguana iguana</i> (46.2%, –74%), <i>Python regius</i> (14.4%, +29%), <i>Testudo horsfieldii</i> (3.7%, +1935%), <i>Varanus exanthematicus</i> (2.8%, –25%), <i>Cuora amboinensis</i> (2.7%, –83%), <i>Boa constrictor</i> (1.9%, –69%), <i>Crocodylus niloticus</i> (1.4%, +6091%), <i>Varanus salvator</i> (1.3%, +233%), <i>Stigmochelys pardalis</i> (1.1%, +375%), <i>Podocnemis unifilis</i> (1.2%, +90775%), <i>Chamaeleo senegalensis</i> (1.1%, +21%), <i>Amyda cartilaginea</i> (1.1%, –34%), <i>Testudo graeca</i> (1.0%, +296%), <i>Varanus niloticus</i> (0.9%, –72%), <i>Python bivittatus</i> (0.7%, –88%)
Captive-bred	9622228 (566013 $\pm$ 27482)	<i>I. iguana</i> (82.3%, –69%), <i>B. constrictor</i> (3.3%, –67%), <i>S. pardalis</i> (1.6%, +15167%), <i>P. regius</i> (1.6%, –1%), <i>Testudo graeca</i> (1.3%, +730%), <i>P. bivittatus</i> (1.1%, –83%), <i>Chamaeleo calyptratus</i> (1.0%, +13513%), <i>Chelonoidis carbonaria</i> (1.0%, +1565%), <i>Geochelone sulcata</i> (0.8%, +25585%), <i>Ptyas mucosus</i> (0.8%, +478%), <i>Testudo hermanni</i> (0.7%, +478%), <i>Morelia viridis</i> (0.4%, +1767%), <i>C. niloticus</i> (0.4%, +83233%), <i>Tupinambis teguixin</i> (0.3%, +18%), <i>Geochelone elegans</i> (0.3%, +74425%)
Ranched	3214908 (189112 $\pm$ 23833)	<i>P. regius</i> (63.1%, +3342%), <i>C. niloticus</i> (6.6%, +471%), <i>P. unifilis</i> (5.9%, +39431%), <i>V. exanthematicus</i> (5.2%, +1191%), <i>T. horsfieldii</i> (4.4%, +1390%), <i>C. senegalensis</i> (3.9%, +12022%), <i>V. niloticus</i> (2.8%, +281%), <i>Chamaeleo gracilis</i> (1.3%, +37742%), <i>Heosemys grandis</i> (1.3%, –100%), <i>Kinixys belliana</i> (0.8%, +1583%), <i>Kinixys homeana</i> (0.8%, +3164%), <i>Heosemys annandalii</i> (0.8%, +336%), <i>V. salvator</i> (0.7%, +133%), <i>Python sebae</i> (0.3%, +45%), <i>T. graeca</i> (0.3%, +517%)
Wild	5412285 (318370 $\pm$ 26235)	<i>I. iguana</i> (11.9%, –91%), <i>C. amboinensis</i> (9.3%, –83%), <i>P. regius</i> (9.2%, –97%), <i>T. horsfieldii</i> (8.2%, +1034%), <i>V. exanthematicus</i> (6.1%, –61%), <i>V. salvator</i> (4.1%, +233%), <i>A. cartilaginea</i> (3.7%, –34%), <i>Uromastix dispar</i> (2.2%, –61%), <i>Naja naja</i> (1.9%, –97%), <i>Phelsuma laticauda</i> (1.7%, –91%), <i>Cordylus tropidosternum</i> (1.7%, –96%), <i>Chamaeleo dilepis</i> (1.7%, –72%), <i>C. senegalensis</i> (1.4%, –81%), <i>V. niloticus</i> (1.4%, –99%), <i>Phelsuma lineata</i> (1.3%, –89%)

3.2.1. Captive-bred reptiles

The trade in captive-bred species expanded from 77 to 152 species in 2012 (16 of which were listed post-1996). Trade of captive-bred *I. iguana* decreased from 93% to 58% of captive-bred imports, whilst trade in other species, notably Nile crocodile (*Crocodylus niloticus*), leopard tortoise (*Stigmochelys pardalis*) and Indian star tortoise (*Geochelone elegans*) increased over the time period (Table 1).

3.2.2. Ranched reptiles

The trade in ranched species expanded from eight in 1996, to 23 in 2012 (four of which were listed post-1996). Total trade in ranched reptiles represented by *P. regius* decreased from 60% to 42%, whilst the proportion of imports represented by various other species, particularly yellow-spotted river turtle (*Podocnemis unifilis*), graceful chameleon (*Chamaeleo gracilis*) and Senegal chameleon (*C. senegalensis*) increased (Table 1).

3.2.3. Wild reptiles

Imports of wild species changed from 113 in 1996, to 121 in 2012, representing an overall decrease considering that 20 of the species traded in 2012 were listed post-1996. *I. iguana* was the most heavily traded wild reptile (Table 1), but decreased from 29% to 9% from 1996 to 2012. Imports in some wild species increased over the time period, most notably Horsfield's tortoise (*Testudo horsfieldii*), whilst imports in many others decreased (Table 1).

3.3. Regional trends

3.3.1. Main reptile exporting regions

The proportion of live reptiles exported from range countries was significantly higher in 1996 (range countries represented 98% of trade) than in 2012 (range countries represented 92% of trade) (Sign test: $Z = 2.04$, $p < 0.05$), when tested across 91 top traded species (100 most traded species excluding nine species listed post-1996).

Considering total imports from 1996 to 2012 on a regional scale, Mesoamerica was the largest exporting region, closely followed by sub-Saharan Africa. South America was the third largest exporting region, followed by South and South-East Asia, and West and Cen-

tral Asia (Fig. 3). See Appendix (Table A2) for a breakdown of trade levels and species exported from each region.

There was evidence of shifting regional patterns over time. This included a rapid drop in imports from South America in the late 1990's to an all-time low in 2010, before a rise in 2011 and 2012. Meanwhile imports from Mesoamerica increased from 1996 to a peak in 2004 and then decreased. Imports from West and Central Asia and North America steadily increased over the time period, whilst imports from sub-Saharan Africa and South and South-east Asia have fluctuated (Appendix: Fig. A3).

3.3.2. Regional trends in captive-bred reptiles

Exports in captive-bred reptiles were dominated by Mesoamerica and South America (Fig. 4, Appendix: Fig. A3(b)). This was largely due to *I. iguana*, in which trade shifted from South America (mainly Colombia) to Mesoamerica (mainly El Salvador) in the late 1990s. Imports of captive-bred reptiles from Sub-Saharan Africa have fluctuated, with a notable increase in imports of captive-bred leopard tortoises (*S. pardalis*): 151 in 1996 to 24656 in 2012. There has been a steady increase in imports of captive-bred reptiles from North America over the time period; notably *P. regius* which increased from zero in 1996 to 12414 in 2012, supplying 8% of the total trade in this species in 2012.

3.3.3. Regional trends in ranched reptiles

Sub-Saharan Africa dominated exports in ranched reptiles, increasing rapidly from 6323 individuals in 1996 to 265936 in 2007, with *P. regius* accounting for 74% of this trade. However, by 2010 trade in ranched reptiles from this region had decreased by 55%. Meanwhile there was a sharp increase in ranched reptiles originating from South America, South and South-East Asia, and more gradually from West and Central Asia (Fig. 4, Appendix: Fig. A3(c)). Imports of ranched reptiles from South and South-East Asia have been sporadic, increasing suddenly in 2010. This mainly included giant Asian pond turtle (*Heosemys grandis*, 38% of trade in ranched reptiles from this region), yellow-headed temple turtle (*H. annandalii*, 24%) and common water monitor (*Varanus salvator*, 22%). The more recent increase from South America was largely due to imports of ranched yellow-spotted river turtle (*P. unifilis*,

96%). The increase in imports from West and Central Asia was predominantly due to trade of ranched *T. horsfieldii* (91%).

3.3.4. Regional trends in wild sourced reptiles

Wild trade decreased in many regions including sub-Saharan Africa, South America and Mesoamerica (Fig. 4, Appendix: Fig. A3(d)). However, there was a slight increase from West and Central Asia consisting mainly of *T. horsfieldii* (86%). Trade in wild reptiles from South and South-East Asia has fluctuated greatly.

3.3.5. Main reptile importing regions

The United States was the principal importing country accounting for over half (56.1%) of all live reptile imports (Fig. 3(b)). The

EU was the second largest importer (18.2%), followed by Mexico (5.8%) and Hong Kong (5.8%) (Appendix: Table A1). Imports into the United States have decreased by 76.8% from 1996 to 2012, whilst imports into Europe have decreased by 44.2% from 2006 to 2012.

4. Discussion

According to the data, international trade in CITES Appendix II live reptiles has decreased over the last ten years, with a dramatic switch from wild sourcing to ranching. Equally, there has been a small increase in the proportion of reptiles sourced from non-range countries. Despite this, the diversity of species represented in trade

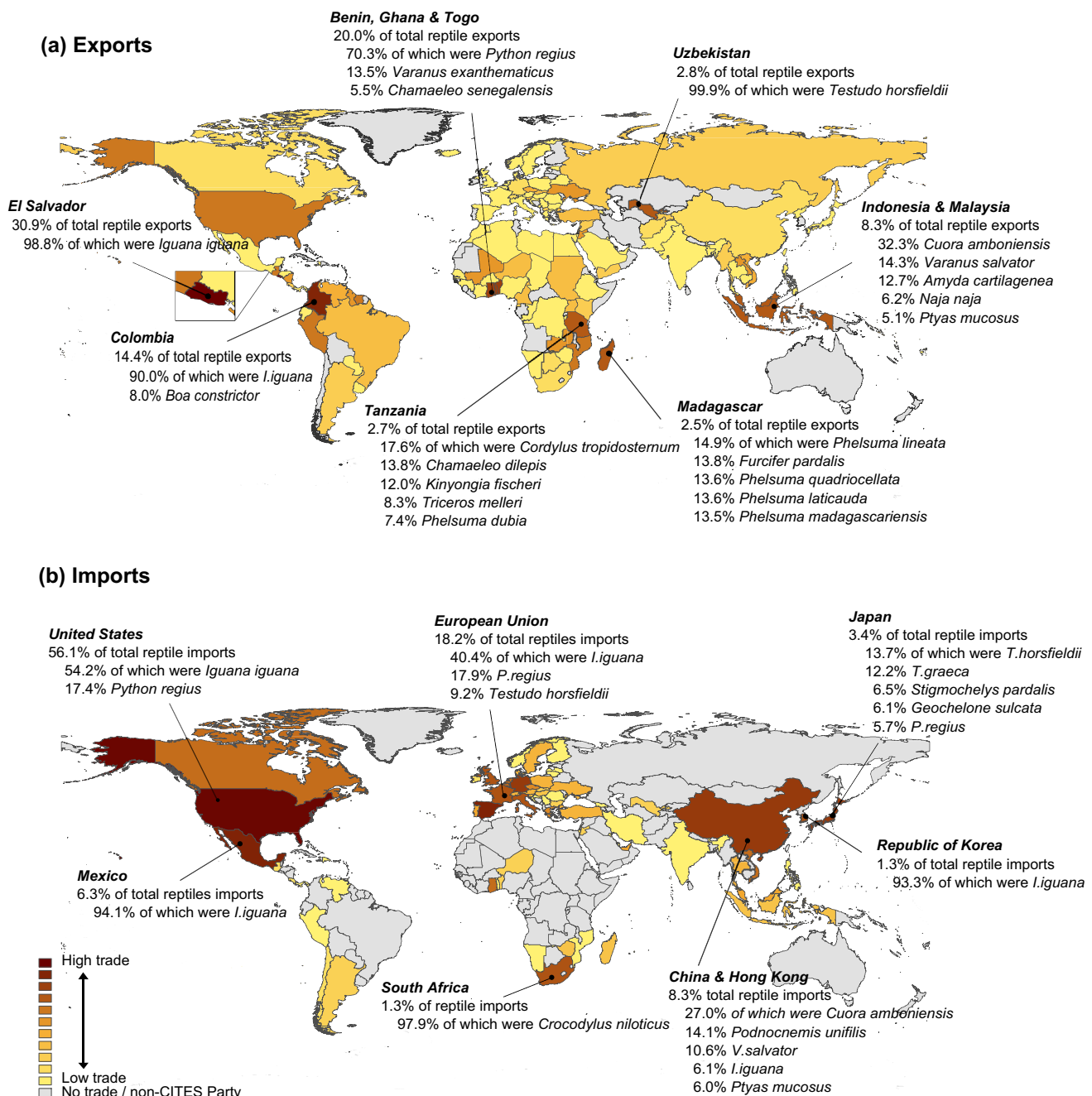


Fig. 3. World map displaying (a) levels of exports and (b) levels of imports in live CITES Appendix II reptiles traded by individual countries for commercial purposes from 1996 to 2012. Figures based on importer-reported quantities obtained from the CITES Trade Database. Countries with text are those with the highest trade volumes. Countries from similar geographical areas have been pooled for the annotations e.g. 'Ghana, Benin and Togo', 'Malaysia and Indonesia'; however more information on exporting countries and regions is available in Appendix: Tables A1 and A2.

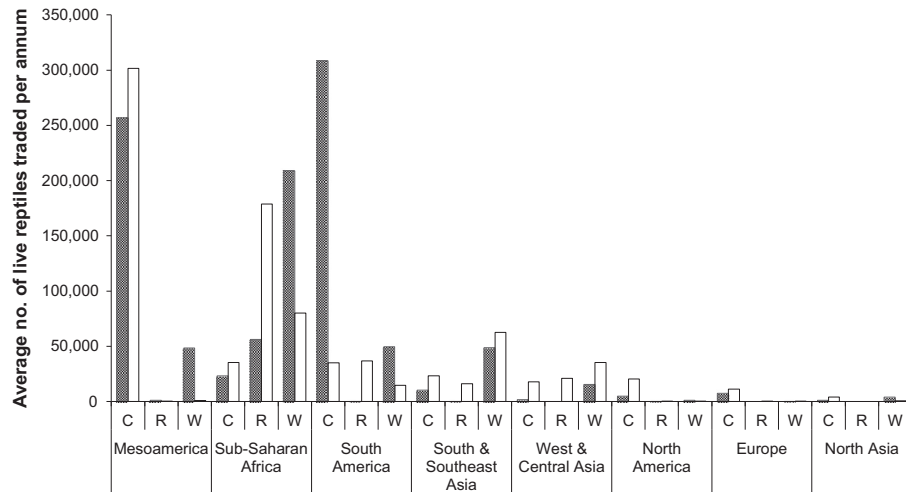


Fig. 4. The average number of live CITES Appendix II reptiles traded for commercial purposes as captive-bred (C), ranched (R), and wild (W) for two five year periods: 1996–2000 (grey bars) and 2008–2012 (white bars). Oceania, East Asia, North Africa and the Caribbean Islands export on average less than 2000 individuals per year and are omitted from this figure.

appears to have increased, reflecting the growing demand for a wide variety of species (Auliya, 2003; Stuart et al., 2006).

Our analysis focussed on direct trade only to reduce double counting of re-exported specimens, allowing better representation of the true numbers and sources of individuals in live trade. However, importing countries may not represent the final destination of specimens. For example, the US plays a major role as a re-exporter of imported animals, probably due to its well-established pet market and central location (Hoover, 1998; Schlaepfer et al., 2005). Additionally, import data do not always represent the final use for specimens. For example, much of the trade in live Nile crocodiles (*C. niloticus* – seventh most traded species) is likely to be supplying crocodile farming industry for skins and/or meat. Additionally, trade in South Asian box turtle (*Cuora amboinensis* – fifth most traded species), is thought to supply demand for consumption and Traditional Chinese Medicine (TCM) (Schoppe, 2009). It is assumed that much of the remaining trade is indicative of trade destined for pet markets.

It is important to note that the data analysed comprise a subset of the overall global trade in reptiles as much trade goes unrecorded (e.g. illegal, domestic, and non-CITES trade). Additionally, this analysis does not include trade in products and derivatives (e.g. skins). Comparisons have been drawn between CITES data and national trade data for the US using the USFWS CITES Enforcement Management Information System (LEMIS), which includes non-CITES trade. Herrel and van der Meijden (2014) reported a positive relationship ($r^2 = 0.71$) between the total import number of live reptiles and amphibians recorded by the USFWS and the CITES database between 2000 and 2009. They also reported a number of other similarities between datasets, including the approximate proportion of imports comprised of captive bred individuals and a recent decline in total imports. However, there were notable differences in the species dominating trade with *Trachemys scripta* (non-CITES) comprising 77% of reptiles traded, and for exports, the CITES data were a less reliable indicator of trade from the US. Schlaepfer et al. (2005) made similar comparisons for 1998–2002 and reported that 28–56% of the 25 most commonly traded reptile species in the US were CITES listed (including pets, skin products, food and TCM). It is difficult to determine how indicative of overall global wildlife trade CITES data are, therefore, findings drawn from data presented herein should not be taken to represent all trade in reptiles. Nonetheless, CITES data form the only comprehensive dataset of global reptile trade and therefore provide an invaluable

source of information for monitoring emerging trends in an important subset of species considered threatened by international trade. For a detailed explanation of limitations and caveats associated with CITES trade data see UNEP-WCMC (2012).

There are a variety of complex factors and interactions affecting trade dynamics including CITES controls (new species listings on appendices, trade suspensions, quotas), regional measures (e.g. European import suspensions and opinions), national measures (domestic legislation, trade policies and national quotas), taxonomic changes, biological effects (species rarity), as well as market and economic forces. For these reasons, providing an in-depth explanation of causal factors for specific trade patterns, across several species and countries at the same time is not viable. A number of studies have provided region or taxa specific analyses of trade data (Arroyo-Quiroz et al., 2007; Carpenter et al., 2005; Li and Jiang, 2014; Luiselli et al., 2012) but, as far as we are aware this is the most comprehensive global overview of the commercial trade in live Appendix II reptiles.

4.1. Decline in global reptile trade

The decline in commercial trade of Appendix II live reptiles may be due to a number of factors, specifically including increased legislative controls, reduced demand, increased captive breeding within consumer nations or external factors such as the global economic climate. The reduction in trade in *I. iguana* was a contributing factor as this species accounted for nearly half of all trade and its imports have decreased considerably, possibly due to reduced demand (Stephen et al., 2011). However, this species is not entirely responsible for the reduction in global imports. Furthermore, a total of 54 species imported over the time period have been newly listed on the appendices since 1996, including a number of turtle species from South-East Asia which feature heavily in trade. This includes *Cuora* spp. (*C. amboinensis* – fifth most traded species) at CoP11 (Kenya, 2000), *Heosemys* spp. (*H. grandis*, *H. annandalii* and *H. spinosa*) at CoP12 (Chile, 2002) and *Amyda cartilaginea* (12th most traded species) at CoP14 (Bangkok, 2004). Therefore the trade data show an overall decreasing trend since 2001, despite additional species listings.

Increased captive breeding of reptiles outside of their source countries may be satisfying an increasingly large proportion of the demand for pet reptiles within consumer nations, thereby reducing demand for international trade. Data to this effect are

lacking, but, there is a significant increase in the percentage of reptiles exported from non-range countries in 2012 compared to 1996. This reflects trade in captive-bred animals, often occurring in developed, consumer nations where adequate resources and expertise exist. Despite the proportion of global trade represented by captive-bred individuals remaining relatively constant between 1996 and 2012, trade in captive-bred *I. iguana* decreased considerably, indicating that numerous other species are increasingly being imported as captive-bred. Additionally, when *I. iguana* is removed from the dataset, there is evidence of an increase in imports of captive-bred reptiles over the time period. Furthermore, Herrel and van der Meijden (2014) reported a steadily increasing proportion of captive bred individuals in the trade when analysing USFWS export data from the US.

The benefits and impacts of captive breeding for supplying commercial trade in wildlife are subject to much debate (Abbott and van Kooten, 2011; Bulte and Damania, 2005; Drury, 2009; Hutton et al., 2001). On one hand, captive breeding reduces pressure on wild populations and particularly for reptiles, consumers often prefer captive-bred specimens as they are easier to keep owing to lower aggression and reduced parasitic infection (Auliya, 2003). However, wildlife trade can bring considerable foreign capital to countries of origin, with benefits passed to a network of hunters and collectors (Auliya, 2003; Roe, 2002). In certain circumstances sustainable trade may support livelihoods of collectors, which can create values for traded species, and promote their conservation, whilst also increasing political motivation to direct resources towards conservation (Hutton and Webb, 2003). However, if the market is saturated by captive-bred specimens from beyond source countries, trade in wild specimens may be displaced thus reducing benefits returning to stakeholders in range countries, which may in turn threaten incentive-based conservation. Even when practiced within range countries, captive breeding may benefit few people and have limited positive impacts on conservation (Arroyo-Quiroz et al., 2007; Hutton and Webb, 2003). Essentially all *I. iguana* in international trade are produced on a few large commercial farms that could produce enough iguanas to exceed world market demand – small farms do not appear to be economically viable and the conservation value of iguana farming is thought 'dubious' (Stephen et al., 2011). Captive breeding within range states can also negatively affect species conservation as it may rely on harvest of wild animals to bolster breeding stock (Haitao et al., 2007) and could create a loophole for illegal laundering of wild animals (Lyons and Natusch, 2011). For this reason Madagascar currently does not allow exports of captive-bred CITES reptiles and amphibians (Madagascar Management Authority, personal communication).

4.2. Shifting production systems – increase in ranching

Whilst the international trade in CITES Appendix II wild and captive-bred reptiles decreased between 2001 and 2012, there has been a substantial increase in imports of ranched reptiles. A negative correlation between ranched and wild caught reptiles does not signify causation, but it may indicate that trade in ranched reptiles is replacing some trade in wild reptiles. Although the trade in ranched reptiles is dominated by *P. regius*, when this species is removed, the relationship remains, demonstrating that numerous other species are increasingly being traded as ranched. A similar switch in trade from wild to captive-bred and ranched sources is well known for crocodiles (Hutton et al., 2001) and has also been demonstrated for monitor lizards (Pernetta, 2009).

The shift towards more highly managed and concentrated systems such as captive breeding, artificial propagation and ranching is often associated with increased trade controls (Jenkins and Broad, 1994; Roe, 2002). This is promoted within CITES through

exemptions for captive-bred Appendices I and II specimens, and a resolution allowing transfer of populations from Appendix I to Appendix II specifically for ranching, provided the programme benefits conservation of the local population. Ranching was originally used for crocodilians transferred from Appendix I to Appendix II and has been instrumental in improving the conservation status of some crocodilian species (Ross, 1998; Thorbjarnarson and Velasco, 1999) whilst also displacing illegal trade (Hutton and Webb, 2003) and generating funds for surveys and management (Jenkins and Broad, 1994; Thorbjarnarson, 1999). Over the years, ranching has been increasingly applied to exports in Appendix II species, but there is little known about the systems employed. In the case of *P. regius* from Ghana, gravid females are collected from wild and kept in captivity until they have laid their eggs which are then artificially incubated. Following this, females plus 10% of neonates are re-released back into the wild (Gorzula et al., 1997). *P. regius* is collected from traditionally managed farmland by villagers who are said to protect the snakes as a key resource (Luiselli et al., 2012). A similar practice is used for ranching of *V. exanthematicus* (Bennett, 2001) and Giant Amazon River Turtles *Podocnemis expansa* in the Amazon (Bonach et al., 2003).

Given that captive breeding is often economically unviable in developing countries (Lyons and Natusch, 2011), ranching may provide a more cost-effective means to supply trade. As well as reducing impacts on local populations relative to wild collection, ranching systems have potential to benefit livelihoods and promote incentive-based conservation if local people are involved in collection of young life stages from the wild. However, undoubtedly a switch from wild collecting to ranching would affect some beneficiaries (Roe, 2002) and there are limited baseline data on the contribution that reptile trade makes to livelihoods in developing countries. Furthermore, the benefits of such systems are dependent upon complex systems of resource rights; if hunters do not own the land where animals are collected, they may not be empowered to protect it (Auliya, 2003). Additionally, the impacts concerning genetics, survival and disease of re-released animals are not well understood.

Unfortunately, the use of source code 'R' (indicating ranched animals) in CITES has been subject to some ambiguity. Following CoP14 (Netherlands, 2007) its use was reviewed and concluded that it had been used erroneously in several countries. Despite guidance provided by CITES on the use of source codes, there are likely to be inconsistencies amongst data which is reported by Parties. The review acknowledged that some ranching operations seemed to have some conservation benefit (CoP15, Doc 29), but there were also indications that benefits may be absent, unknown or questionable. However, very little information was received with only six out of 27 Parties involved in the review responding. Given that significantly more reptiles are apparently being sourced from ranching operations it is imperative that more research is directed towards understanding the benefits and impacts of such systems for commercially traded reptiles.

5. Conclusion

Commercial trade in live wildlife is dynamic and complicated by a variety of factors. Increased legislative control, economics or consumer demand may have driven shifting production of reptiles from wild to ranched sources. Previous research has shown that trade bans may not always be the best solution for controlling wildlife trade (Rivalan et al., 2007). Indeed, sustainable and well-managed wildlife collection has potential to support livelihoods and promote valuation and stewardship of species and habitats (Andreone et al., 2006). In this case the substantial increase in ranched reptiles in place of wild reptiles may have significant

conservation and sustainability consequences, as pressure is reduced on wild populations and benefits from the trade are retained within range states. Management of the trade in order to ensure sustainability remains a challenge, but certification schemes such as the FSC Forest Certification Scheme play an important role in other traded products. It remains to be seen whether such systems could be applied to the exotic pet trade, stimulating sufficient demand from end-users.

In terms of conservation, the best mode of supplying live reptiles for trade will vary depending on the species conservation status, biology and demand. Ranching programs are based on utilising young life stages that would normally have high mortality in the wild, and are therefore applied to species with relatively large clutch sizes and high mortality between the egg and adult life stages. Additionally, whilst species with high population estimates and a broad distribution could withstand certain levels of exploitation associated with wild trade or ranching, collection of species with low population estimates could easily lead to local extinction (Andreone et al., 2006). In addition to this, due to the boom-and-bust nature of wildlife trade, programs based predominantly on the sale of wildlife products for a single market makes conservation outputs vulnerable so for programs to ensure sustainability, they should find ways to diversify outputs (Thorbjarnarson, 1999).

Knowledge of consumer demand and preferences are undoubtedly important in understanding future dynamics of the live reptile trade. Although some consumers may prefer captive-bred animals, there are likely to be others who seek rare individuals not currently available in captivity, or prefer wild sourced animals in order to bolster bloodlines. Therefore, without a good understanding of the demand for traded species, it is difficult to know whether programs based on breeding commercially important species will reduce demand for their wild counterparts. A better understanding of the impacts of alternative modes of production, such as ranching, along with an improved understanding of consumer demand in the commercial live trade may help us understand not only the drivers of change, but also mitigate for negative consequences of the trade in the future.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.biocon.2014.12.019>.

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